

In this note we have some simple Markdown examples and show their conversion to PDF via LaTeX.

## Handling sections

Use H1 (#) for the title, and then H2 (##) for sections. H2 headers start at the **subsection** level, and are not numbered. H3 headers (###) and lower just get turned into **paragraphs**.

### A subsection

Here's a subsection.

**A subsubsection** ... is now a **paragraph** environment.

## Equations

Here's an inline equation:  $E = mc^2$ .

Here's a regular equation:

$$-\frac{\hbar^2}{2m}\nabla^2\psi(r,t) + V(r)\psi(r,t) = i\hbar\frac{\partial}{\partial t}\psi(r,t) \quad (1)$$

For multi-line equations, you can use the **align** environment (or **gather**, or **eqnarray**, etc...) like this:

$$a = b + c \quad (2)$$

$$x = y - z \quad (3)$$

Unfortunately the raw LaTeX doesn't render in most Markdown previewers, e.g. VSCode. However, for reasons that are mysterious to me, the VSCode Markdown previewer does allow for newlines via `\\` in equation mode. So if you want to do multiline like that to start, and then switch to the **align** environment manually before compiling to LaTeX/pdf, go for it.

Note that you don't need to add a blank line before a full-line equation in the Markdown file, e.g. you can write

```
some text
```

```
$$
```

```
1+1=2
```

```
$$
```

instead of

```
some text
```

```
$$
```

```
1+1=2
```

```
$$
```

And here's a matrix:

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}. \quad (4)$$

## Inline code

Here's some `inline code`.

## Lists

Here's a simple list:

- x
- y
- z

Here's an enumerated list:

1. x
2. y
3. z

Here's an enumerated list with an equation:

1. x
2. y
3. an equation:

$$a + b \quad (5)$$

4. z

## Tables

Here's a simple table. Something is a little off with the rendering, not sure why...

| Syntax | Description |
|--------|-------------|
| Header | Title       |

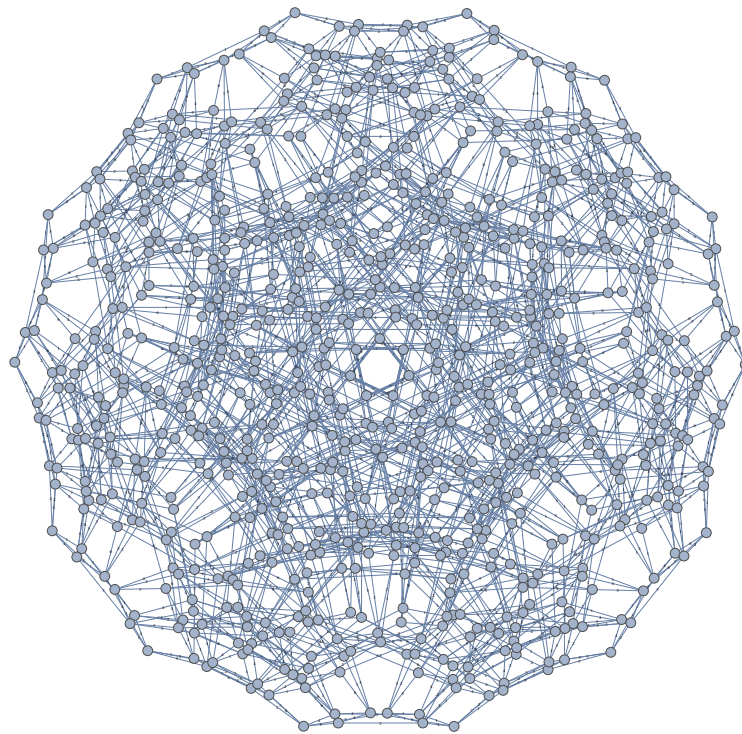


Figure 1: A caption

Paragraph    Text

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## Figures

Here's a figure with a caption: